

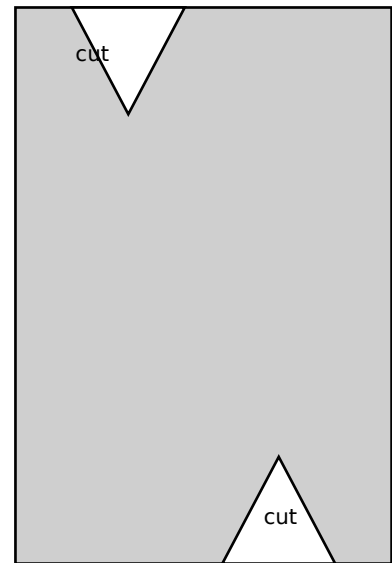
Set 5 - Mixed Exam-style

Level: Mixed Challenge

Name: _____ Class: _____ Date: _____

Page 1

1. Sharon has a rectangular paper measuring 60 cm by 30 cm. She cuts five similar triangles from the top. Each triangle has base 12 cm and height 15 cm. Find the area of the remaining paper.

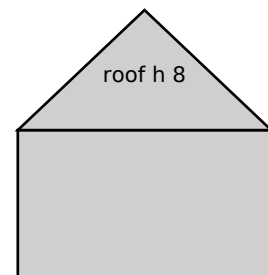


60 x 30; 5 cuts, each 12 x 15

Working:

Answer:

2. A house-shaped figure is made of a rectangle 18 cm by 14 cm and a triangular roof with base 18 cm and height 8 cm. Find the total area.



18 x 14

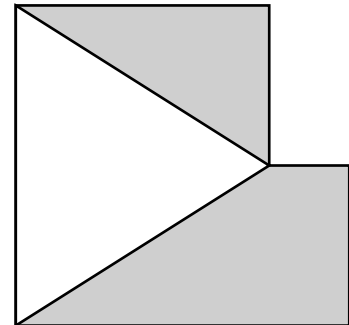
Working:

Answer:

Name: _____ Class: _____ Date: _____

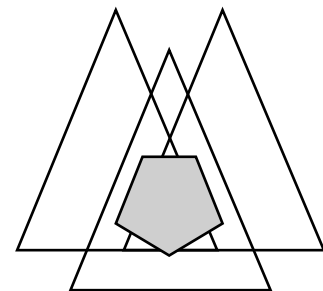
Page 2

3. An L-shaped figure is made from a 36 cm by 24 cm rectangle by removing a 12 cm by 8 cm rectangle. Find the area of the L-shaped figure.

 $36 \times 24 \text{ minus } 12 \times 8$ **Working:**

Answer:

4. Three identical triangles overlap. Each triangle has base 30 cm and height 20 cm. The central shaded overlap has area 75 cm^2 and is counted in all 3 triangles. Find the area covered by the whole figure.

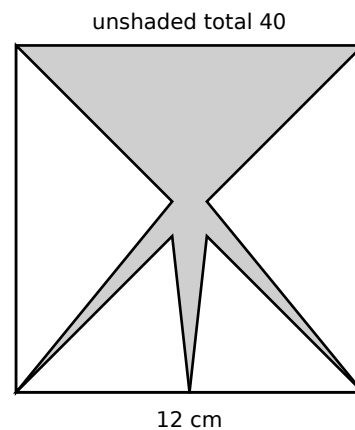


3 triangles; overlap 75

Working:

Answer: _____

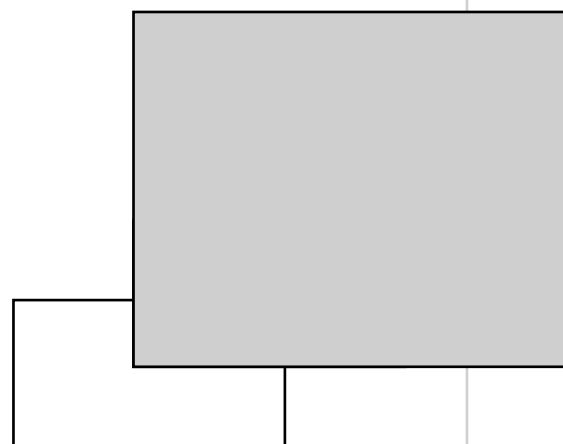
5. A square paper measures 12 cm by 12 cm. The total area of the unshaded triangles is 40 cm². Find the shaded area.



Working:

Answer:

6. Two identical rectangles each measure 20 cm by 8 cm. They overlap by 48 cm². Find the area of the combined figure.



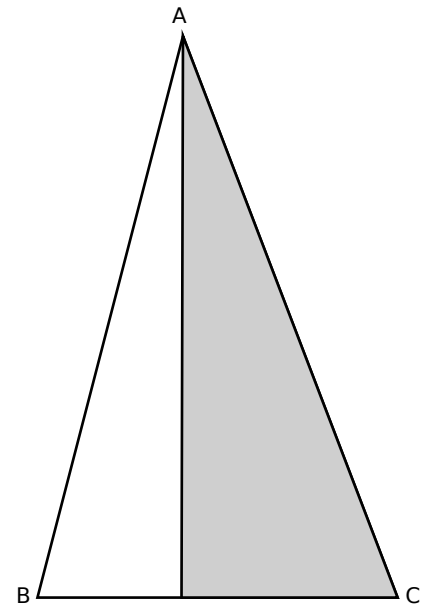
Working:

Answer:

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7. In Triangle ABC, D lies on BC. $BD = 10$ cm and $DC = 8$ cm. The area of Triangle ADC is 32 cm^2 . Find the area of Triangle ABC.

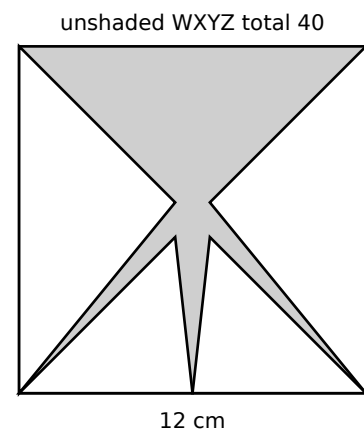


$BD=10$, $DC=8$
same height

Working:

Answer:

8. PQRS is a square with side 12 cm. The total area of the unshaded triangles W, X, Y and Z is 40 cm^2 . Find the area of the shaded part.



Working:

Answer: _____

Set 5 - Mixed Exam-style

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- 9.** Ethan spent $\frac{2}{7}$ of his money on a bag and $\frac{3}{5}$ of the remainder on shoes. He had \$80 left. How much money did he have at first?

Working:

Answer:

- 10.** Zoe baked some muffins. She gave $\frac{1}{4}$ of them to her neighbours and sold $\frac{1}{3}$ of the remainder. She had 30 muffins left. How many muffins did she bake at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	1350 cm ²	Rectangle = 1800. One triangle = 90. Five = 450. Remaining = 1350 cm ²
2	324 cm ²	Rectangle = 252; roof = $\frac{1}{2} \times 18 \times 8 = 72$; total = 324 cm ²
3	768 cm ²	$36 \times 24 - 12 \times 8 = 864 - 96 = 768 \text{ cm}^2$
4	750 cm ²	Total counted = $3 \times 300 = 900$. Overlap counted 3 times but needed once, subtract 2×75 Area = 750 cm ²
5	104 cm ²	Square area = 144; shaded = $144 - 40 = 104 \text{ cm}^2$
6	272 cm ²	$2 \times 160 - 48 = 272 \text{ cm}^2$
7	72 cm ²	Same height. $BC/DC = 18/8$. Area ABC = $32 \times 18/8 = 72 \text{ cm}^2$
8	104 cm ²	Square = 144; shaded = $144 - 40 = 104 \text{ cm}^2$
9	\$280	Left = $\frac{5}{7} \times \frac{2}{5} = \frac{2}{7}$ of total. $\frac{2}{7} = 80$, total = 280
10	60 muffins	Left = $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ of total. Total = 60